TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dyeing of Different Fibres with Mordant Dyes and Pigments

Coordination lakes on cotton and wool with aluminium, chromium and iron — and a binder-fixed pigment dyeing that colours a blend no dye could level.

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 3 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to dye cotton and wool with a **mordant dye** using metal mordants, and to colour cotton and polyester/cotton with **pigments**, comparing shade, handle and fastness behaviour. A mordant dye (e.g. alizarin type) has chelating groups (–OH/–COOH in ortho positions) that coordinate a metal ion; the metal simultaneously binds fibre sites, forming an insoluble coloured **lake** anchored by coordination bonds. The metal is part of the colour: Cr^{3+} gives deep, dull, very fast shades; Al^{3+} bright reds (Turkey red tradition); Fe^{2+}/Fe^{3+} darkens and saddens shades towards brown/black.

Fibre ⋯ M^{n+} ⋯ Dye (coordination lake, insoluble, formed in/on the fibre)

2. Apparatus and materials

- Lots: cotton 5.0 g ×3, wool 5.0 g ×1, polyester/cotton 5.0 g ×1; LR 1:30 (150 mL baths).
- Mordants: potash alum $KAl(SO_4)_2 \cdot 12H_2O$ 10% owf (0.50 g), potassium dichromate $K_2Cr_2O_7$ 3% owf (0.15 g — Cr(VI): handle as hazardous, gloves, collect waste), ferrous sulphate $FeSO_4$ 5% owf (0.25 g).
- Mordant dye 2% owf (0.100 g via stock); pigment dispersion 3% owf equivalent with binder (Section 4).

3. Procedure — mordant dyeing (pre-mordanting method)

1. **Mordanting:** dissolve the mordant in the bath, enter pre-wetted fabric at 40 °C, raise to 90–95 °C (wool) / boil (cotton, with 5 g/L soda ash assistant for alum), hold 45 min. Rinse lightly — the metal must stay on the fibre.
2. **Dyeing:** fresh bath with dye stock (and 2% owf acetic acid for the chrome bath on wool), enter mordanted fabric at 40 °C, raise slowly to 95 °C, hold 45 min. The lake forms progressively —

shade deepens and shifts as chelation completes.

3. **After-chrome variant (wool standard):** dye first with the mordant dye, then treat with dichromate 1–2% owf at 95 °C for 30 min in the same bath — compare its shade and fastness with the pre-mordanted piece.
4. Rinse, soap lightly, dry. Record shade per metal: same dye, three metals, three colours — the demonstration that the metal is a co-chromophore.

4. Procedure — pigment dyeing of the blend

1. Pad-style in the lab: prepare pigment liquor — pigment 20 g/L, binder 40 g/L, antimigrant 10 g/L. Immerse the polyester/cotton lot, squeeze to ~70% pick-up (weigh before/after to verify: target wet mass = 5.0 × 1.70 = 8.5 g).
2. Dry at 100 °C, cure 150 °C for 3 min. Pigment add-on = 20 g/L × 0.70 L/kg = 1.4% owf (Topic 4 arithmetic).
3. Compare blend levelness with the mordant cotton: the pigment colours both fibres identically; no dye in this experiment could.

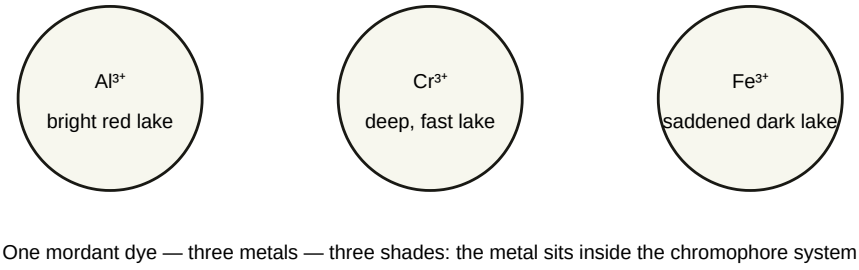


Fig. 1 — Mordant metal as co-chromophore: aluminium, chromium and iron lakes of one dye.

5. Results (as recorded)

SAMPLE	SHADE RESULT	HANDLE / NOTES
Alum mordant, cotton	Bright red	Soft; moderate wash fastness
Chrome mordant, wool	Deep bordeaux	Best fastness of the set (tested in Experiment 5)
Iron mordant, cotton	Dark maroon-brown	Saddening unmistakable
Pigment, P/C blend	Level mid-blue	Slightly firmer handle; colour on both fibres equally

6. Precautions

- Dichromate is toxic and an environmental hazard: minimum quantities, gloves, dedicated waste container — and the reason chrome mordanting is restricted in modern production despite its fastness.
- Iron mordant in excess tenders cotton on storage; rinse thoroughly.
- Verify pigment pick-up by weighing — guessing pick-up falsifies the add-on calculation.

7. Practice (viva preparation)

1. Draw/describe the coordination structure that makes a lake insoluble.
2. Why does iron "sadden" a shade while aluminium brightens it?
3. Pre-mordanting versus after-chroming: which gives better fibre protection for wool, and why?
4. Calculate pigment add-on if pick-up were 80% at the same 20 g/L liquor.

Answer key

1. The metal ion accepts lone pairs from dye oxygens/nitrogens (chelate rings) while also coordinating to fibre donor groups; the crosslinked metal–dye–fibre assembly is a large, charge-neutral, insoluble complex — a lake — mechanically and chemically locked to the fibre.
2. Iron's d-electron transitions add broad absorption across the visible (and Fe–dye charge-transfer bands), removing light in several regions at once → darker, duller. Aluminium has no d electrons/absorption of its own; the dye's own chromophore shows cleanly → bright.
3. After-chroming — the wool is dyed under mild acid conditions first and meets Cr(VI) only briefly at the end, limiting chromium uptake by and damage to the fibre; pre-mordanting loads metal through the whole dyeing.
4. 20 × 0.80/10 = 1.6% owf.
